Cloud Platforms (AWS)

Session 1: Cloud and AWS Foundations

MIBA24 · Master of Science in Business Analytics

ESADE Business School | February 4, 2026

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Show Attendance QR Code

Students: scan with your phone to register attendance

Open eCampus

Oriol Rius

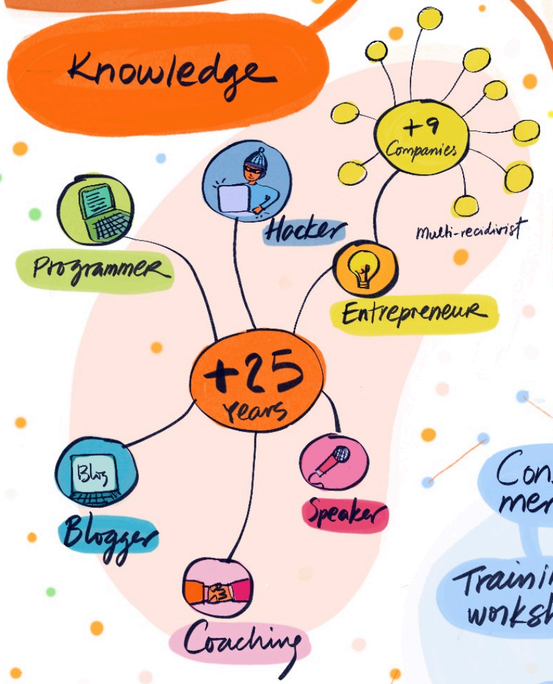
"To lead is to serve"

Purpose: "Help to built a world where people can live on their passion!"

Sharing

"Working in different cultures allows you to have BIG IDEAS"

Knowledge



Experience

I offer

Consultancy & mentoring

Trainings & workshops

Presentations

TEDx

Industry 4.0

Humanizing Technologist of the 4th Industrial Revolution

I help to

overcome technical CHALLENGES (Solutions Architect)

Make teams flow like Water

Humanize the technologists

Develop trainings and strategies for managers and sales

Creativity

"Life is not about meeting, it is about creating yourself."

Expert in teams without office

"I know I'm leading when I create a trail of leaders along the way"

Lean & Agile from ever since

accredited adviser ACCIÓ

"Life is growth. To continue growing, I now have to share."

"DON'T FIGHT, JUST FLOWS"

oriolrius.me

VISUAL STORYTELLING by ALEYANDRA ETEL
db INTERSECTION.COM

€ Quick Math

Netflix + HBO + Spotify ≈ **€35/month**

Raise your hand if you pay for streaming

ChatGPT Pro or Claude Pro ≈ **€20/month**

Keep it up if you also pay for an AI tool

Where are you investing in your future?

Today's Agenda

Course Overview

Cloud and AWS Foundations

The 3Rs

What these words mean **in this classroom**

Respect

Embrace diverse backgrounds and perspectives

Listen actively — let others finish

Debate openly — challenge ideas, not people

Be present — phones away during class

Contribute responsibly to group dynamics

Responsibility

80% attendance mandatory (QR code)

Arrive prepared

Arrive on time

Check eCampus regularly

Miss a quiz = grade 0 (no make-ups)

Justified absences → Program Manager

Results

Learn by breaking things

Build real pipelines

Fail fast, learn faster

Deliver real outcomes

Have fun doing it

> “The only stupid question is the one that remains unasked

How We Work Together

Punctuality

- Class starts **exactly** at scheduled time
- Door closes when class begins — no late entry
- Late arrival = absence for that session

During Class

- Silence during explanations
- Quick comment to a classmate? **OK**
- Side conversations? **Not OK**
- No leaving class — use toilet before or during break

⚠ Disruptive behavior → expelled from class → counts as absence

Part 1: Course Overview

Cloud + Data + ML = Scalable Business Analytics

Cloud Platforms (AWS) — The 9-Session Journey

3 ECTS · ~75 hours total workload

FOUNDATIONS

Session 1
Cloud & AWS Foundations

Session 2
SQL Fundamentals Schema & CRUD

Session 3
Advanced SQL Joins & Triggers

AWS Console & CLI · EC2 & S3 · RDS PostgreSQL · SQL Design Patterns

Feb 4 → Feb 18, 2026

CONTAINERS & ML

Session 4
Docker, ECR & ECS Fargate

Session 5
SageMaker I Training & Registry

Session 6
SageMaker II Pipelines & Deploy

Dockerfile · ECR Push · ECS Tasks · SageMaker Training · Model Registry · Pipelines

Feb 25 → Mar 18, 2026

INTEGRATION & ASSESSMENT

Session 7
Building APIs Gateway + Lambda

Session 8
Project Work & Presentations

Session 9
FINAL EXAM Theory + Hands-on

API Gateway · Lambda Functions · End-to-End Pipeline · CloudWatch Monitoring

Mar 25 → Apr 14, 2026

Cloud & SQL · Containers · Machine Learning · APIs & Integration · Assessment

Final Project: AWS Analytics Pipeline (30%)

Learning Objectives

1. Apply Cloud Principles

Cloud-computing principles to business-analytics workflows
AWS infrastructure: compute, storage, networking, IAM

2. Design Data Pipelines

Relational data pipelines in the cloud
SQL, AWS RDS, CRUD, joins, triggers, views

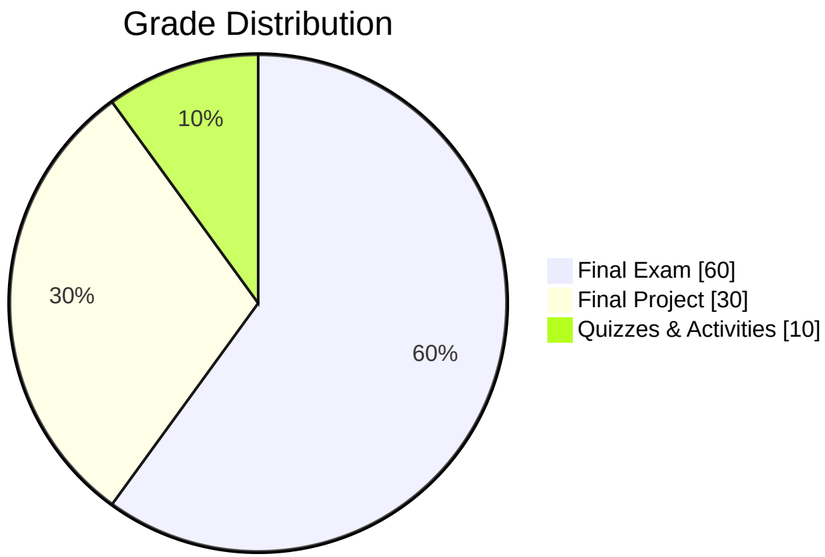
3. Containerize Workloads

Containerize and orchestrate analytical workloads
Docker, Amazon ECR, ECS Fargate

4. Train & Deploy ML

Machine-learning models using AWS SageMaker
Model registry, pipelines, inference endpoints

Assessment Overview



Final Exam

60%

Theory + hands-on (SQL, Docker, AWS) | **Min 4/10 to pass**

Final Project

30%

AWS-based analytics pipeline with RDS, ECS, SageMaker & API

Quizzes & Activities

10%

Session-bound checks, mini-labs, design reviews

The Final Project 30%

A reproducible AWS-based analytics pipeline deployed on cloud infrastructure

Technical Components

- **RDS + SQL** — Schema design & analytical queries
- **S3** — Data storage and versioning
- **Docker/ECR** — Container images
- **ECS Fargate** — Container deployment
- **SageMaker** — Model training & registry
- **API Gateway/Lambda** — Inference delivery

Deliverables

- **Git repo** — Clean history, tags/releases
- **README** — Architecture, setup, execution steps
- **SQL scripts** — Schema and analytical queries
- **IaC config** — CloudFormation or equivalent
- **Deployment evidence** — Screenshots, CLI logs
- **Demo** — 5-8 min recording or live

Evaluation Rubric

40%	25%	20%	15%
Implementation & Reproducibility	Architecture & Security	Documentation & Traceability	Cost Efficiency & Monitoring

What I Expect From You

Attendance & Punctuality

- **80% minimum** to pass (max 2 absences)
- Door closes when class begins — no late entry
- Late arrival = absence for that session
- Register via **QR code** each session

Preparation

- Review pre-class materials
- Tools installed & working (AWS, Docker, SQL client)
- Check eCampus regularly
- Try before asking

Mindset

"Learning by building and analyzing"

- Experiment with architectures
- Debug configurations
- Evaluate analytical output
- Iterate and improve

Communication

- Office hours: Wed 14:30-16:00
- No weekend/late-night emails
- Questions already in syllabus/eCampus won't be answered

Workload & Tools

AI Coding Assistants Policy

AI Tools Are ALLOWED

GitHub Copilot, ChatGPT, Claude, Cursor, etc.

1. Cite Usage

Document in your README which AI tools you used and for what purpose

2. Verify & Adapt

Test all outputs, fix errors, adapt code to your specific needs

3. Understand It

You must be able to explain every line of code you submit

AI-generated content without understanding = Honour Code Violation

Submitting code you cannot explain will be treated as plagiarism

Part 2: Cloud Foundations

Why cloud computing matters for business analytics

Before We Start

What comes to mind when you hear **"cloud computing"**?

The Internet?

Storage?

Servers?

Cost savings?

Cloud = Someone else's computer that you rent by the minute

The IT Challenge & Solution

Why are companies moving to the cloud?

Traditional IT

The Pain Points

- **Capital Expense** — $50K-500K+ upfront
- **Procurement** — 3-6 months wait
- **Capacity Planning** — Guess 3-5 years out
- **Underutilization** — 15-20% average use
- **Maintenance** — Cooling, power, patches
- **Staff** — Full IT team required

Cloud Computing

The Solution

- **Pay-as-you-go** — Only pay what you use
- **Instant** — Servers in seconds
- **Elastic** — Auto-scale up/down
- **Global** — Deploy anywhere
- **Managed** — Provider maintains
- **Reliable** — 99.99% uptime SLAs

Key shift: From CapEx to OpEx — From guessing to responding — You can't buy a server in 10 minutes, but you can rent one in 10 seconds

Cloud Service Models

How much do you want to manage vs. delegate?

IaaS

Infrastructure as a Service

- Applications
- Data
- Runtime
- Middleware
- OS
- Virtualization
- Servers
- Storage
- Networking

Ex: EC2, Azure VMs

PaaS

Platform as a Service

- Applications
- Data
- Runtime
- Middleware
- OS
- Virtualization
- Servers
- Storage
- Networking

Ex: Lambda, ECS, Heroku

SaaS

Software as a Service

- Applications
- Data
- Runtime
- Middleware
- OS
- Virtualization
- Servers
- Storage
- Networking

Ex: Gmail, Salesforce, O365

You Manage | Provider Manages

This course focuses on **PaaS** services

Why AWS?

Market Leader

32%
Market Share

33
Regions

105
Avail. Zones

200+
Services

Why for This Course?

- **Industry standard** — most jobs
- **Best ML services** — SageMaker
- **AWS Educate** — free credits

Cloud Market Share

Transferable Skills

Skills transfer to **any** cloud:

Concept	AWS	Azure	GCP
Compute	EC2	VMs	GCE
Storage	S3	Blob	GCS
Containers	ECS	AKS	GKE

Bottom line: AWS skills make you employable everywhere — and transfer to Azure/GCP when needed

15-Minute Break

Stretch, grab coffee, check messages

Back at **10:45** sharp

AWS Core Services for Analytics

Service	Category	What It Does	Course Usage
S3	Storage	Object storage (files, data)	Data lake, model artifacts
RDS	Database	Managed relational databases	SQL analytics, data storage
EC2	Compute	Virtual machines	General compute
Lambda	Compute	Serverless functions	Event-driven processing
ECS/ECR	Containers	Container orchestration	Deploy ML services
SageMaker	ML/AI	End-to-end ML platform	Train, deploy, monitor models
API Gateway	Networking	REST/HTTP APIs	Expose ML endpoints
CloudWatch	Monitoring	Logs, metrics, alarms	Monitor everything

By end of course, you'll have used **all of these** in your project

AWS Global Infrastructure

Regions 33

Physical geographic locations worldwide

- `eu-west-1` Ireland
- `us-east-1` Virginia
- Data stays in region (GDPR)
- Choose by latency & compliance

Availability Zones 105

Isolated data centers within a region

- Multiple AZs per region
- Low-latency interconnects
- Deploy across AZs for HA
- Fault isolation by design

Edge Locations 400+

Content delivery closest to users

- CloudFront CDN caching
- Route 53 DNS resolution
- Reduced latency globally
- Closest to end users

Critical Reminder Important

Check your region before creating resources

- Always verify region in console
- Resources are region-specific
- ! Wrong region = "not found" error
- **Course region: eu-west-1**

For this course: We use `eu-west-1` (Ireland) — always check top-right corner of AWS Console before creating any resource

AWS Shared Responsibility Model

Who is responsible for what?

Security OF the Cloud

AWS Responsibility

- Physical data centers
- Network infrastructure
- Hypervisor/virtualization
- Hardware maintenance
- Managed service operations
- Global infrastructure

Security IN the Cloud

Your Responsibility

- Data encryption
- IAM & access control
- Network configuration (VPC)
- Application security
- Operating system patches
- Firewall rules

Key Principle: You control WHO accesses WHAT and HOW — if your S3 bucket leaks data, that's YOUR fault

AWS Identity & Access Management (IAM)

Core Concepts

Users
Individual identities (people, apps)

Groups
Collections of users (dev-team, analysts)

Policies
JSON documents defining permissions

Roles
Temporary credentials for services

Best Practices

- **Never** use root account
- **Enable** MFA everywhere
- **Follow** least privilege
- **Use** roles, not keys
- **Rotate** credentials
- **Audit** with CloudTrail

Policy Example

```
{
  "Effect": "Allow",
  "Action": "s3:GetObject",
  "Resource": "arn:aws:s3:::my-bucket/*"
}
```

Only allows reading from one bucket

AWS Pricing Model

Pay-as-you-go

Only pay what you use

EC2: Per second

S3: Per GB

Lambda: Per request

Reserved Capacity

Commit 1-3 years, save 72%

Best for: Predictable

Example: Prod DB

Savings: Up to 72%

Free Tier

12-month trial + always-free

Lambda: 1M req/mo

S3: 5 GB

RDS: 750 hrs

Set Alarms!

- ✓ Create **$10 budget alert**
- ✓ Stop unused EC2 instances
- ✓ Delete resources after labs
- ✓ AWS Innovation Sandbox credits are limited

Live Demo: AWS Console

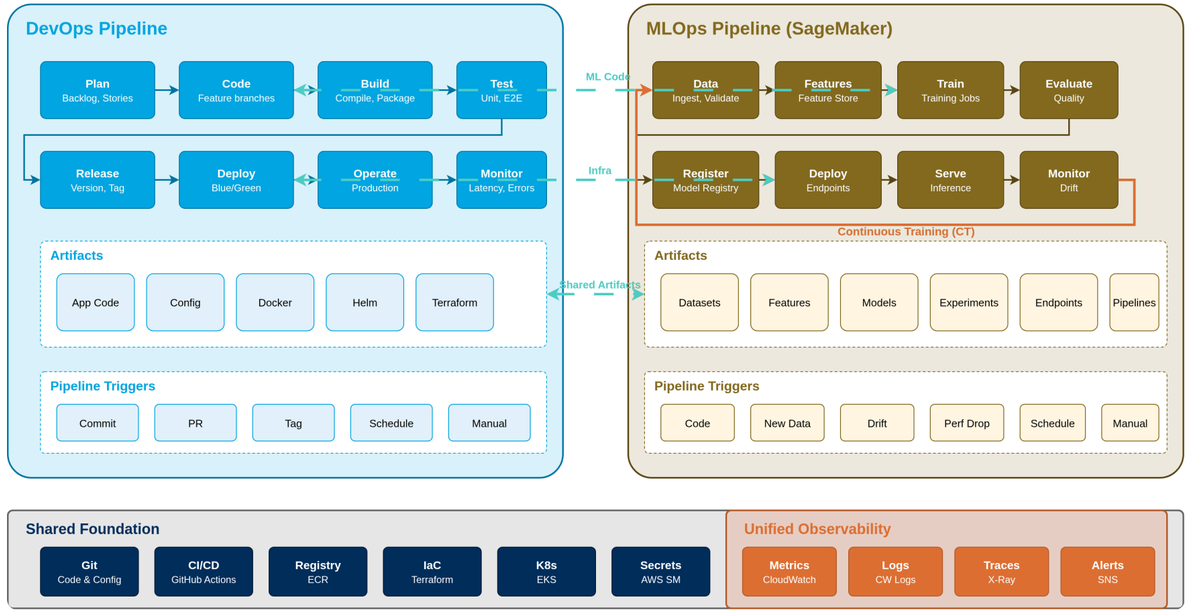
1. AWS Innovation Sandbox request
2. Wait for approval
3. Budget control
4. Access AWS Console from Sandbox
5. Region Selection `eu-west-1`
6. AWS CLI configuration
7. AWS CLI access
8. AWS EC2 usage
9. SSH access to EC2
10. AWS S3 upload object

Cloud for Business Analytics

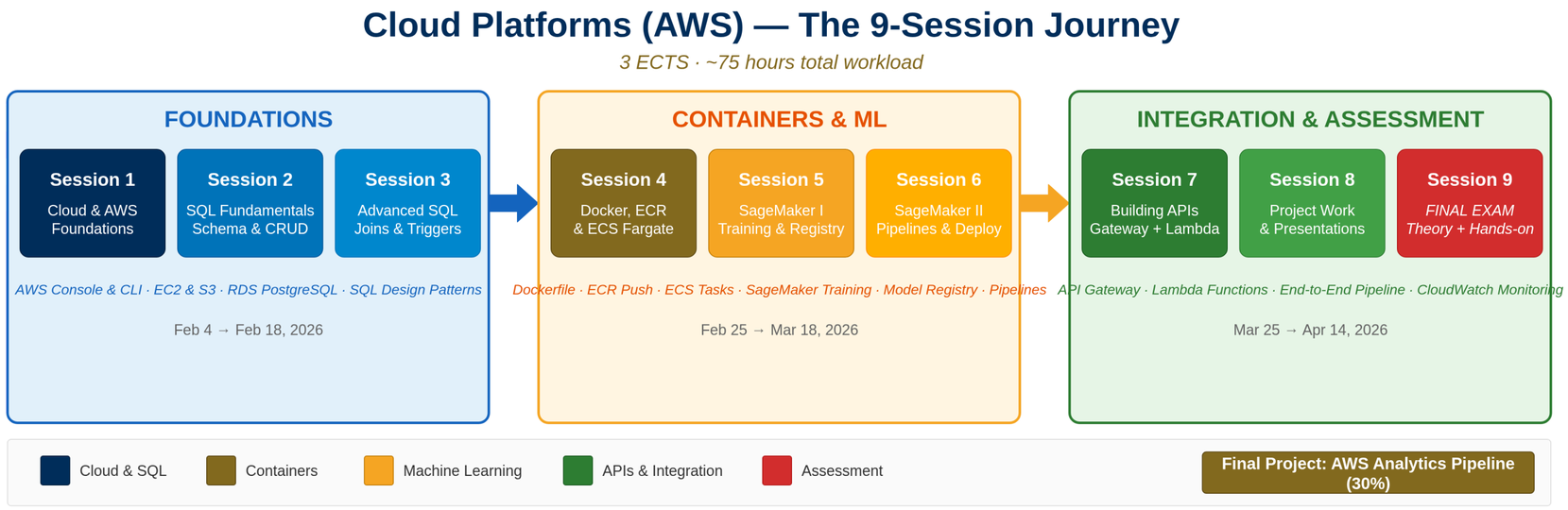
The pipeline you'll build in your final project

1. Ingest	2. Process	3. Train	4. Deploy	5. Insights
S3 + RDS	Lambda/Glue	SageMaker	ECS + API GW	Dashboards

DevOps + MLOps: Unified Platform Architecture



Your Pipeline Journey



Sessions 1-3
Cloud + SQL/RDS
Data foundations

Sessions 4-6
Docker + SageMaker
ML capabilities

Sessions 7-8
APIs + Project
Integration

Session 9
Final Exam
Theory + Hands-on

Progressive learning: Each session builds on the previous — by Session 8, you'll have a complete AWS analytics pipeline ready for the exam

Key Takeaways

Cloud = Renting Infrastructure
Pay-as-you-go, instant scaling, global reach. No upfront hardware investment.

Costs Add Up Fast
Set billing alarms NOW. Stop unused resources. Free tier has limits.

AWS Dominates the Market
32% market share, 200+ services. Skills transfer to Azure/GCP.

Region Matters
Always check your region. We use **eu-west-1** (Ireland) for this course.

Shared Responsibility
AWS secures the cloud. YOU secure what's in it. IAM is your gatekeeper.

Core Services for Analytics
S3, RDS, Lambda, ECS, SageMaker, API Gateway. You'll use them all.

Homework

Repeat the live demo steps on your own before next session

1. **AWS Innovation Sandbox request**
2. **Wait for approval**
3. **Budget control**
4. **Access AWS Console from Sandbox**
5. **Region Selection** `eu-west-1`
6. **AWS CLI configuration**
7. **AWS CLI access**
8. **AWS EC2 usage**
9. **SSH access to EC2**
10. **AWS S3 upload object**

⚠ **Important:** Complete steps 1-3 immediately — sandbox approval may take time!

Questions?

Now is the time to ask anything about:

- **Course logistics**
- **Cloud concepts**
- **AWS services**
- **Final project**

Office hours: **Wednesday 14:30-16:00**

Or email for appointment (not questions already in syllabus)

Thank You!

See you next class!

Questions? → joseporiol.rius@esade.edu